

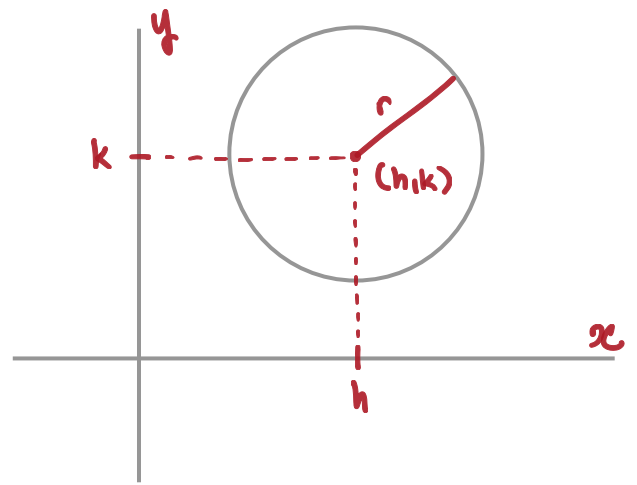
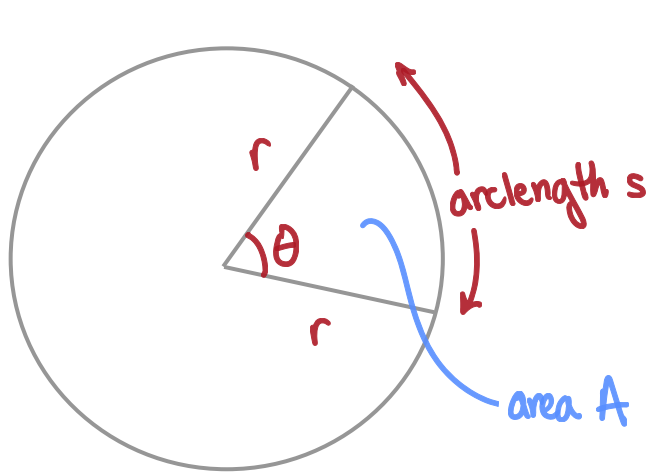
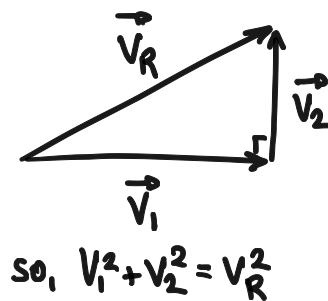
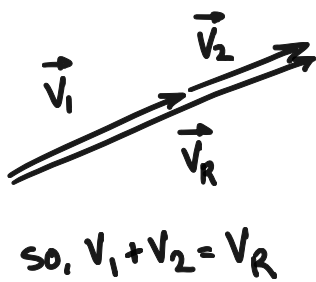
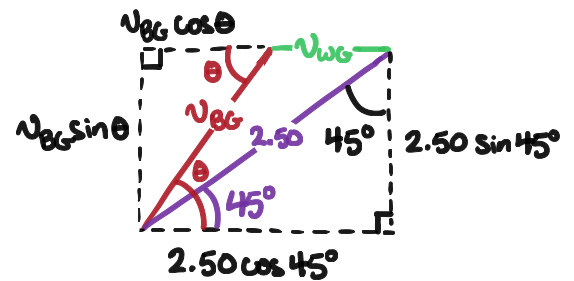
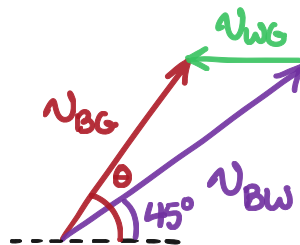
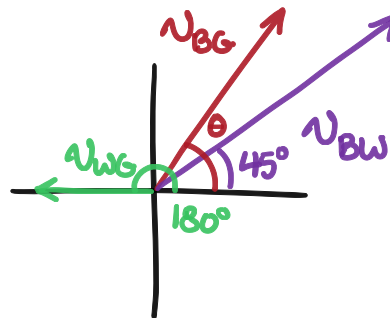
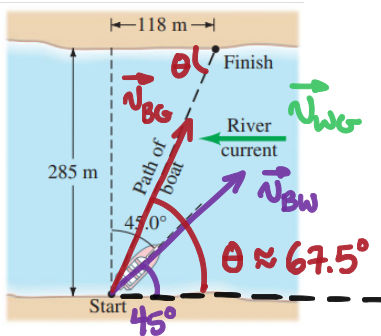
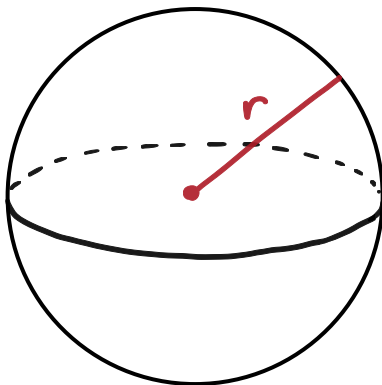
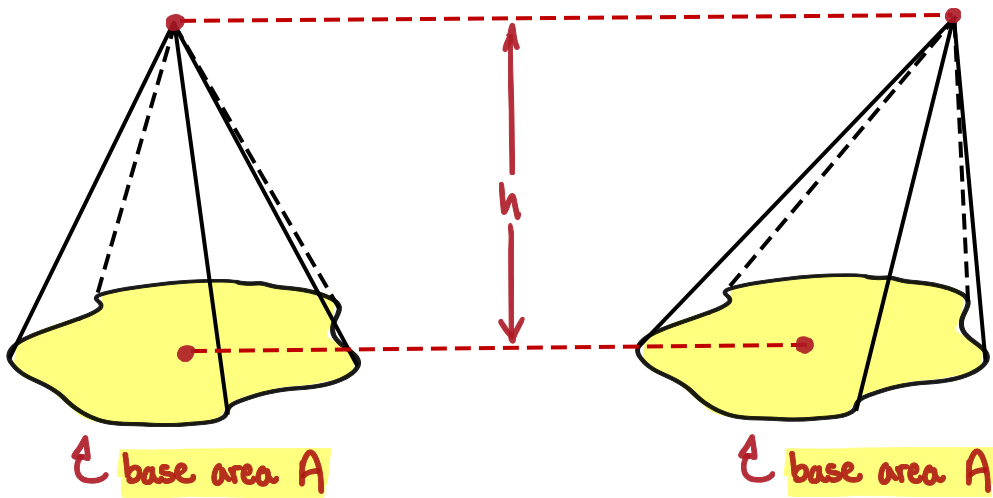
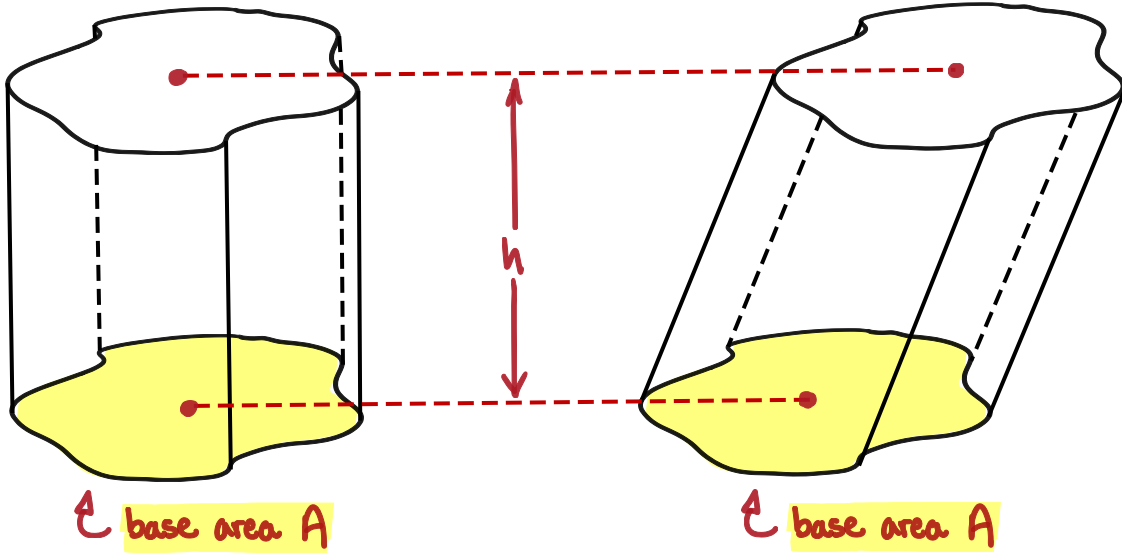
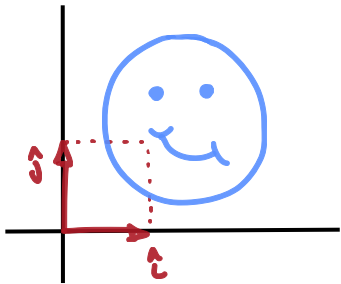
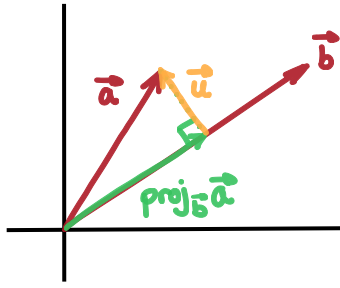
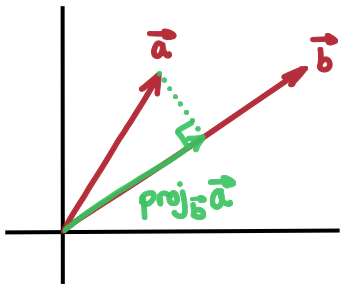


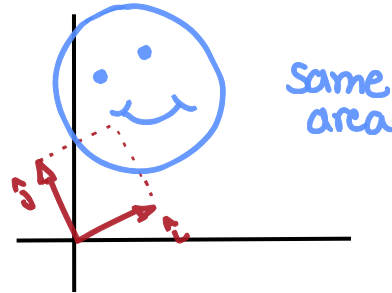
FIGURE 3-45
Problem 48.





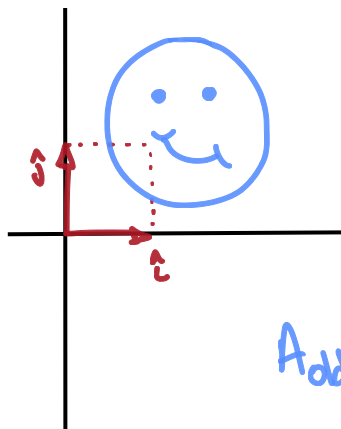


M
rotation

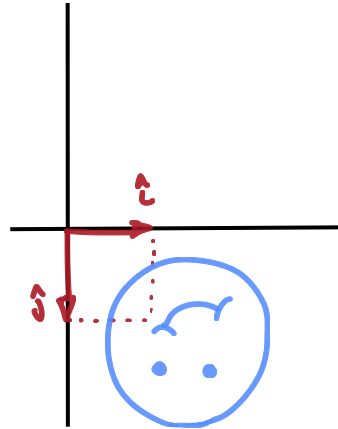


$$A_{\text{old}} \cdot 1 = A_{\text{new}}$$

$\uparrow \det M = 1$



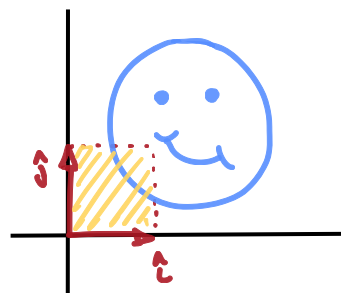
M
x-axis
reflection



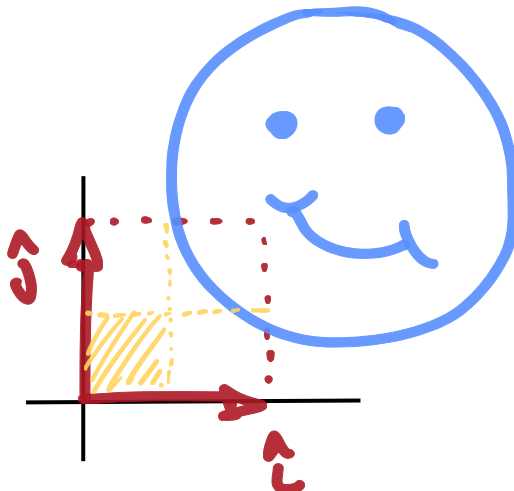
$$A_{\text{old}} \cdot 1 = A_{\text{new}}$$

$\uparrow \det M = -1$

(minus sign indicates flipped image)

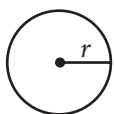
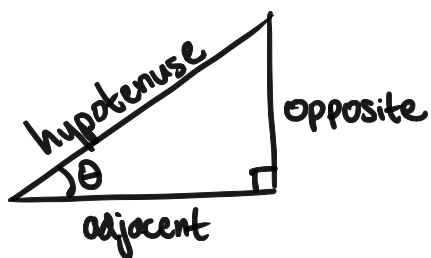


M
 $\times 2$ in x
and y directions



$$A_{\text{old}} \cdot 4 = A_{\text{new}}$$

$\uparrow \det M = 4$

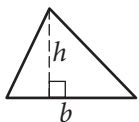


$$A = \pi r^2$$

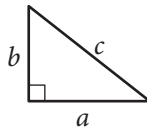
$$C = 2\pi r$$



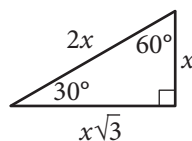
$$A = \ell w$$



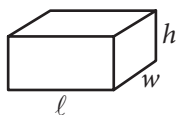
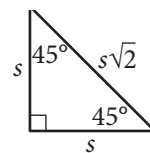
$$A = \frac{1}{2}bh$$



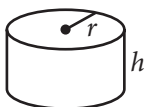
$$c^2 = a^2 + b^2$$



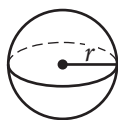
Special Right Triangles



$$V = \ell wh$$



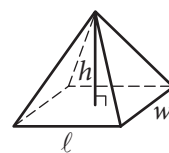
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

